Event-Resolved Gravimetric Detection of Typhoon-Driven Ocean Mass Redistribution

[PENDING: author list and data-provider collaboration]

Methods-first draft; not for submission

Abstract

We test whether named typhoons produce event-resolved ocean mass-redistribution signals in superconducting-gravimeter records after explicit separation of atmosphere, hydrology, direct ocean attraction, elastic loading, and instrument effects. The design combines gravimetry with pressure, best tracks, gridded sea level, tide gauges and GNSS, and evaluates a completely held-out typhoon together with strong-weather and quiet controls. [**PENDING: Insert data-gated multi-event results and uncertainty after station authorization.**]

1 Introduction

Previous work has modeled non-tidal ocean loading in the South China Sea and observed storm-surge loading at Helgoland. The increment tested here is not general NTOL correction, but named-typhoon attribution with a closed component budget, independent evidence, and cross-event generalization.

2 Data and preregistered event design

The raw-data gate requires either one station covering at least three typhoons or two stations covering at least two typhoons each. Required inputs include gravity, colocated pressure, calibration and maintenance logs, timing and jump records, sea-level fields, storm tracks, precipitation, and hydrology. Evaluation additionally requires one held-out typhoon, one non-typhoon strong-weather control, three quiet windows, and no overlapping same-station windows across splits.

IBTrACS agency wind and pressure fields are retained separately. ERA5 surface pressure supplies atmospheric loading; mean-sea-level pressure and winds are event diagnostics. Significant wave height is not converted into additional water mass. The frozen CMEMS MOWL product separates dynamic sea level, inverse barometer, tide, steric and mass-volume contributions [1]. We select its `sea_surface_height` component, which excludes tides and surface-pressure forcing, and add the ERA5-driven inverse-barometer response exactly once.

3 Methods

3.1 Transparent gravimeter correction

For observed gravity g_{obs}, the residual is

$$g_{\mathrm{res}} = g_{\mathrm{obs}} - \sum_k g_k, \tag{1}$$

where every removed component has a unique physical-effect identifier, source, SI series, and pre-applied flag. Two components cannot own the same effect. Each stage stores its input, removed series, output, and peak contribution.

Missing samples and cadence violations split records without interpolation. Candidate discontinuities are flagged at original indices but are not automatically classified or corrected.

3.2 Independent ocean forward model

Sea-level anomaly η is converted to surface density $\Delta\sigma = \rho_w \eta$. Direct attraction is integrated on spherical and WGS 84 grids with explicit masks and partial coastal-cell fractions. The modeled ocean signal is

$$g_{\text{ocean}}^{\text{model}} = g_{\text{direct}} + g_{\text{deformation}} + g_{\text{internal}}, \tag{2}$$

with every term retained separately. Elastic terms are disabled until the Green-function Earth model, reference frame, normalization, and component semantics pass a published benchmark.

3.3 Double-count audit

A CMEMS field containing inverse-barometer response cannot be combined with a second ERA5-driven inverse-barometer correction. Likewise, a pre-corrected NTOL residual cannot receive NTOL again. Unknown product physics blocks assembly.

3.4 Evaluation

Primary metrics are correlation, RMSE, peak amplitude and timing error, explained variance, spectral coherence, event SNR, and held-out performance. Quiet and control windows estimate false positives. Sensitivity tests cover the model, filters, hydrology, Green functions, stations, and events; failed events are retained.

4 Results

A registered non-restricted readiness audit connects 17 completed processing and evaluation work units. The open IBTrACS inventory contains 48 broad regional track candidates and 11 priority station-coverage inquiries. These are not gravity events and do not select a pilot: the real-event gate remains `pending_no_event_windows`. A second registered audit closes all six environmental-effect ownership entries. In particular, CMEMS dynamic sea level is the ocean direct/elastic input while the CMEMS inverse-barometer component is excluded in favour of the ERA5-derived branch. This removes a double-counting risk but supplies no gravity observation. Consequently the claim-safe branch is `pending_evidence`. **[PENDING: Insert only registered real-SG event results after the station-data gate passes.]**

5 Discussion

Possible outcomes include successful event attribution, evidence that ocean products miss rapid coastal redistribution, or an upper constraint from non-detection. A single successful event cannot establish generality.

6 Data and code availability

Restricted SG and station logs will remain outside Git. Versioned manifests will record owner, terms, station, channels, time range, product version, retrieval time, checksum, exclusions, and redistribution limits. Processing code and compact nonrestricted metrics will be released.

7 Limitations

SG event windows, calibration and station logs remain authorization-dependent. CMEMS pressure-response semantics and correction ownership are now closed at the metadata level, but event-specific CMEMS/ERA5 retrieval and anomaly construction have not occurred. Consequently no pilot event, attribution coefficient, held-out improvement, event SNR or false-positive rate is reported. The current method audit cannot substitute for those observations.

References

[1] Copernicus Marine Service. Global ocean physics analysis and forecast, 2026. Product GLOBAL_ANALYSISFORECAST_PHY_001_024; PUM and QUID reviewed 2026-07-12.